

ATUL SINGH

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PROFESSIONAL SUMMARY

AI/GenAI Engineer with hands-on experience developing and deploying Generative AI solutions using LLMs (GPT, open-source models), building end-to-end RAG pipelines, prompt engineering workflows, and AI-powered applications. Proficient in Python, FastAPI, LangChain, Azure OpenAI, HuggingFace, and REST APIs. Strong foundation in NLP, ML concepts, and agentic AI orchestration on Azure.

TECHNICAL SKILLS

Core Languages:	Python, SQL
GenAI & LLMs:	LangChain, Prompt Engineering, RAG Pipelines, LLM Fine-tuning, Agentic AI, LangGraph, LlamaIndex
AI APIs & Models:	Azure OpenAI (GPT-4o), OpenAI API, HuggingFace (BAAI/bge-small-en-v1.5, Llama 3.1), Groq
NLP & ML:	NLP, Scikit-learn, Pandas, NumPy, Embeddings, Semantic Search
Vector Databases:	ChromaDB, FAISS, Azure AI Search
Backend & APIs:	FastAPI, REST APIs, JSON, Docker, Git, GitHub
Cloud & DevOps:	Microsoft Azure, Azure Resource Management, Railway, Vercel
Visualization:	Power BI, Matplotlib, Excel

EXPERIENCE

Software Engineer — LTIMindtree, Noida

Oct 2025 – Present

- Develop and implement enterprise-grade Generative AI solutions using LLMs (Azure OpenAI GPT-4o, open-source models via HuggingFace and Groq).
- Build AI-powered applications including intelligent chatbots, and multi-agent workflows using LangChain and FastAPI backends.
- Design and optimize prompt engineering strategies to improve LLM output accuracy, relevance, and latency.
- Integrate AI models using APIs (Azure OpenAI, HuggingFace, OpenAI) into end-to-end GenAI pipelines with vector databases (ChromaDB, FAISS).
- Preprocess and manage unstructured text data for RAG applications; evaluate and monitor model performance across accuracy, latency, and cost metrics.
- Collaborate with data engineers on AI use cases; deploy scalable backend APIs on Azure cloud infrastructure.

Graduate Engineer Trainee — LTIMindtree

Jul 2025 – Oct 2025

- Completed intensive training on Generative AI, LLMs, Agentic AI, LangChain, RAG architecture, vector databases, and NLP concepts.
- Built intelligent AI workflows using industry-standard frameworks; gained hands-on experience with REST APIs, JSON, and cloud-based model hosting.

PROJECTS

1. Codebase Q&A Bot | LangChain · Groq (Llama 3.1) · HuggingFace Embeddings · ChromaDB · FastAPI · React · Docker

- Production-grade AI-powered chatbot that answers natural language questions about any Python codebase — returns exact file names, function names, and line numbers.
- Built AST-based chunking pipeline ensuring retrieved chunks are complete logical units (functions/classes), not arbitrary token splits.
- Integrated BAAI/bge-small-en-v1.5 (HuggingFace) embeddings with ChromaDB for semantic code search; LangChain RAG chain with Groq as free LLM backend.
- Engineered structured prompts that enforce consistent output format (file path, function name, line numbers, plain-English explanation) — core prompt engineering work.
- Containerized with Docker; deployed backend on Railway, React + Tailwind frontend on Vercel. Supports GitHub URL ingestion via GitPython.

Live: <https://codebase-qa-bot.vercel.app> | GitHub: [Atul-Singh-Rajput/codebase-qa-bot](https://github.com/Atul-Singh-Rajput/codebase-qa-bot)

2. Multi-Agent Travel Planner | Microsoft Agent Framework · Azure OpenAI GPT-4o · Streamlit

- Designed and deployed a 9-agent AI system that orchestrates specialized agents (Transport, Lodging, Cultural, Food) in parallel, reducing planning time by 33%.

- Implemented Human-in-the-Loop (HITL) approval stage between pipeline phases, demonstrating enterprise-grade control over high-stakes AI decisions.
- Built shared workflow state management across all agents using WorkflowContext with structured TripProfile and TravelPlan data models.
- Delivered fully interactive Streamlit UI with real-time agent progress monitoring and HTML/PDF itinerary export.

CERTIFICATIONS

- Microsoft Certified: Azure Fundamentals
- GH-300: GitHub Copilot Certification
- Microsoft Azure AI Engineer: Developing ML Pipelines in Microsoft Azure — Pluralsight
- Artificial Intelligence Fundamentals — IBM
- Data Science Fundamentals — University of California, Irvine (Coursera)
- Azure Machine Learning & MLOps: Beginner to Advance — Udemy

EDUCATION

MCA — Artificial Intelligence & Data Science	May 2025
Amrita Vishwa Vidyapeetham, Mysore — CGPA: 9.2	
B.Sc (PMCs)	Sept 2022
SBRR Mahajana First Grade College, Mysore — CGPA: 9.39	